

Biochemistry and Physiology of Marine Animals

Cephalopods' adaptations to their environment : focus on the
Hapalochlaena genus



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1) Origins of the genus *Hapalochlaena*

Cephalopods can be distinguished by their bilateral body symmetry, a well-developed head and a set of arms, which are muscular structures derived from the ancestral molluscan foot. They are currently divided into two main subclasses, which diverged around 430 million years ago: *Nautiloidea* and *Coleoidea* (Whitelaw et al., 2020).

The subclass *Nautiloidea* includes the nautiluses, the only living cephalopods with an external shell. The subclass *Coleoidea* includes two major superorders: the decapodiforms and the octopodiforms. The decapodiforms include cuttlefish, squid and spirulae, all of which have ten appendages (eight arms and two elongated tentacles), as their name suggests.

Octopodiforms have emerged approximately 275 millions years ago and include octopuses and vampyromorphids, for a total of more than 300 species (Figure 1 ; Sanchez et al., 2018 ; Whitelaw et al., 2020 ; Whitelaw et al., 2023).

Concerning the genus *Hapalochlaena*, molecular studies support the monophyletic status, with its most recent divergence from *Octopus bimaculoides* occurring around 59 million years ago. This divergence likely took place after the Eocene-Oligocene extinction, which led to the emergence of new lineages in the Central Indo-Pacific, a place known as a biodiversity hotspot during that period (Sanchez et al., 2018 ; Whitelaw et al., 2023).

Currently, only four species of the genus *Hapalochlaena* have been described and universally accepted: *Hapalochlaena maculosa*, *Hapalochlaena lunulata*, *Hapalochlaena fasciata* and *Hapalochlaena nierstraszi*. These species typically measure between 12 and 20 centimeters in length (Whitelaw et al., 2023 ; Williams & Caldwell, 2024). However, mitochondrial and SNP data analyses suggest that around 11 distinct clades exist, possibly more (Whitelaw et al., 2023).

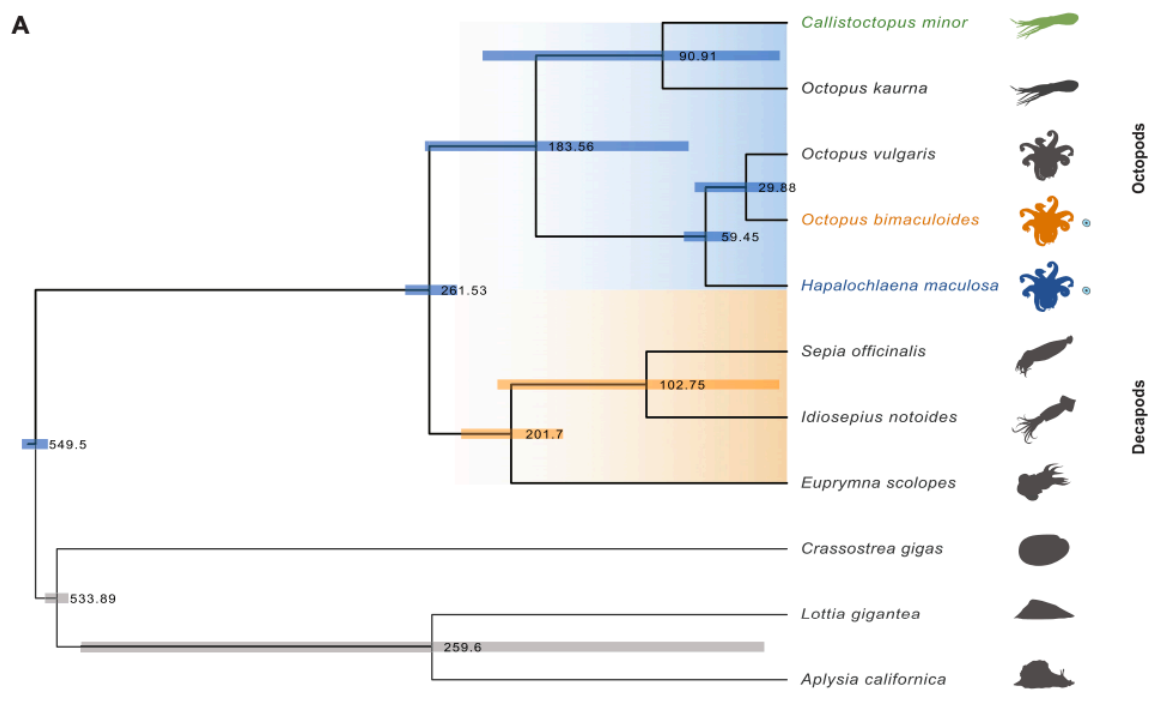


Figure 1: Comparison of molluscan genomes and gene families. (A) A time-calibrated maximum likelihood phylogeny of 7 molluscan genomes (*Aplysia californica*, *Lottia gigantea*, *Crassostrea gigas*, *Euprymna scolopes*, *Octopus bimaculoides*, *Callistoctopus minor*, and *Hapalochlaena maculosa*) and 4 transcriptomes (*Octopus kaurina*, *Octopus vulgaris*, *Sepia officinalis*, and *Idiosepius notoides*), based on 2,108 single-copy orthologous gene clusters. Node labels: divergence times in millions of years (mya) ; blue bars: octopod divergence ; orange bars: decapod divergence. The standard error of those nodes is a 95% confidence interval (Whitelaw et al., 2023).

2) Distribution area

Octopuses can be found in all of Earth's oceans, spanning nearly the entire depth range, from the sea surface to the hadal zone. Their diversity is greatest near the equator and gradually decreases toward the poles (Williams & Caldwell, 2024).

Blue-ringed octopuses from the genus *Hapalochlaena* live in numerous coastal areas in the Pacific and Indian Oceans, ranging from Australia to Japan (Whitelaw et al., 2023 ; Kajino et al., 2024). In these regions, water temperatures can go from 20 to 28°C. These octopuses can be found at depths of up to around 50 meters (Tranter & Augustine, 1973), typically on rocky reefs in shallow waters, mostly in the intertidal and subtidal zones (Williams & Caldwell, 2024).

After storms, blue-ringed octopuses are often found in shallow pools of water, sheltered under rocks, among lightly submerged corals, among colonies of ascidians, or on muddy bottoms rich in algae clusters (Sutherland & Lane, 1969 ; Lambert, 2011).

The species *H. lunulata* is the only one that adopts a merobenthic lifestyle. The other species, *H. maculosa* and *H. fasciata*, are holobenthic, meaning that they spend their entire lives on the ocean floor (Whitelaw et al., 2023).

3) Feeding

Octopuses have a highly varied diet and are opportunistic predators. Their feeding habits depend on several factors, including the size and maturity of the octopus, prey availability, as well as the depth and type of habitat in which they are found (Smith, 2003).

Due to their wide distribution, various behaviors have been observed among octopuses of this genus. *H. fasciata* and *H. maculosa* are considered as temperate species and often take cover under oysters, mussels or other bivalves. Tropical species will most likely shelter in coral rubble. *H. lunulata* has shown diurnal habits while *H. fasciata* is predominantly nocturnal. Regarding their diet, they all feed on crustaceans like crabs, stomatopods and shrimps, as well as gastropods, including mussels (Williams, 2008 ; Williams & Caldwell, 2024).

Hapalochlaena species exploit their “keen eyesight” to ambush and capture preys such as hermit crabs, small crayfish, shrimp, wounded fish and shellfish (Sheumack & Howden, 1978 ; Stewart, 1997). When *Hapalochlaena* attack hard-shelled invertebrates, they use their sharp beaks to perforate their prey's exoskeleton and inject a paralyzing venom

(Geffeney et al., 2019). Using their radula and arms, they then tear into the flesh and suck out the tender, pre-digested tissues of their victim (Lambert, 2011).

4) Reproductive behavior

4.1) Mating behavior

Octopuses display a variety of complex mating behaviors, including changing colors, performing physical displays, and using different mounting techniques. Males often adopt specific body patterns and colors to attract females or establish dominance during interactions with other males (Doucet & Meadows, 2009 ; Mäthger et al., 2009). These displays facilitate sexual recognition and play a key role in both male competition and female mate selection (Doucet & Meadows, 2009 ; Cifuentes et al., 2023).

Blue-ringed octopuses are polygamous, with reproduction occurring in spring and autumn, depending on the species (Lambert, 2011). Reproduction is described as an active process in *H. maculosa*. During mating, the male mounts the female and wraps her tightly around his mantle (**Figure 2** ; Tranter & Augustine, 1973). He then uses a specialized reproductive arm, called a hectocotylus, to transfer spermatophores (sperm packets) to the oviduct, in the female's mantle cavity (Tranter & Augustine, 1973 ; Hanlon & Messenger, 2018 ; Williams & Caldwell, 2024). This transfer process can last several minutes, or even hours, and often takes place in protected areas to minimize predation risk during reproduction (Lambert, 2011 ; Williams & Caldwell, 2024).

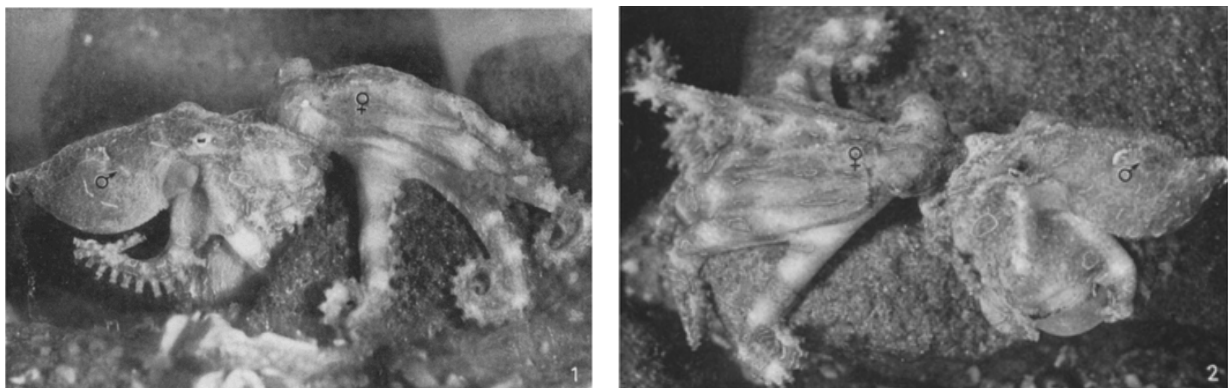


Figure 2 : Copulation of *Hapalochlaena maculosa* aged 4 months (Tranter & Augustine, 1973). (Left) : The male sits on top of the female on her dorsal side, encircling her firmly with his six front arms, while the rear pair of arms rests more loosely. (Right) The third right arm, modified into a hectocotylus, is inserted ventrally into the female's mantle cavity, close to the oviduct openings.

4.2) Social recognition and choice of partner

Within the Cephalopods, octopuses are often described as asocial or even antisocial. Their very specific interactions usually revolve around reproduction. Social recognition is hypothesized to be specialized for facilitating reproduction behaviors such as mate choice and sperm competition. For example, *Abdopus aculeatus* can identify the sex of an individual from a distance due to sex-specific chromatophore patterning (Morse et al., 2022).

Social recognition, which involves distinguishing individuals of the same species, plays a key role in cooperative, reproductive, and competitive behaviors (Doucet & Meadows, 2009). In cephalopods, known for the complexity of their brains and behaviors, this ability is particularly important (Morse et al., 2022). Apart from the common octopus (*Octopus vulgaris*), which displays sexual dimorphism, males of other octopus species cannot easily distinguish between sexes from a distance, despite their well-developed sensory systems (Cheng & Caldwell, 2000).

Studies on *H. maculosa* have shown that females are likely able to discriminate the sex of conspecifics via distance chemoreception using chemical cues. In contrast, males do not appear to have this ability. Additionally, *H. maculosa* has been observed using chemotactile clues to identify socially and biologically relevant traits in the organisms physically in contact (Morse et al., 2022 ; Williams & Caldwell, 2024).

Some retreats behaviours have also been observed, males more often fleeing from female conspecifics than from male ones. Furthermore, male seem to distinguish females they previously mated with and those they have not. However, females showed no use of chemotactile recognition to retreat, which is consistent with previous studies. (Morse et al., 2022 ; Williams & Caldwell, 2024).

Therefore, unlike females, who can use chemical signals in the water to detect sex of conspecific from a distance, males must make physical contact to recognize the female's sex and copulation history (Morse et al., 2022).

Male-male copulation is generally brief, lasting a few seconds to a few minutes, and does not involve the release of spermatophores. This suggests that such interactions may be due to opportunistic or even erroneous behavior, likely due to males' inability to distinguish the sex of their conspecifics. On the other hand, male-female copulation is longer, lasting up to 246 minutes, and involves the release of spermatophores. Unlike male-male interactions, male-female copulation is often aggressively interrupted by the female. Following reproduction, three types of behavior have been observed: avoidance, resumption of copulation or attack. Females occasionally display aggressive behavior towards males after mating. In extreme cases, this interaction can be lethal, for example, a female was documented killing and cannibalizing a male (Cheng & Caldwell., 2000).

Females are able to store sperm from multiple males throughout the breeding season before laying a single clutch with mixed paternity. The copulation chronology of male partners influences paternity rates, particularly when copulation is completed by the female.

Indeed, the first male to mate with a female would have a higher paternity rate if copulation was terminated by the female. Conversely, males mating in second position have higher paternity rates when they terminate the copulation themselves (Morse et al., 2018). Males interrupt copulation more often when females have already mated with other males, due to sperm competition. They strategically adjust copulation duration, investing more time with unmated females and shortening interactions with those already mated. Males thus maximize their reproductive success by modulating their investment according to the probability of paternity (Morse et al., 2015).

Brief flashes of iridescence have sometimes been observed just before mating attempts, both, between males or between males and females. This behavior could play a role in communication between individuals, particularly during reproductive interactions (Mäthger et al., 2012).

4.3) Egg laying

After mating, the female lays between 50 and 100 eggs in a secure area, often within a reef or burrow. She keeps the eggs with her, wrapping them in a “brood pouch” formed by her arms (Tranter & Augustine, 1973 ; Lambert, 2011). This brooding period lasts approximately two months. During this period, the female stops feeding, devoting all her energy to ventilating and cleaning her eggs to ensure they remain oxygenated and protected from infection or predator (Tranter & Augustine, 1973 ; Lambert, 2011 ; Williams & Caldwell, 2024).

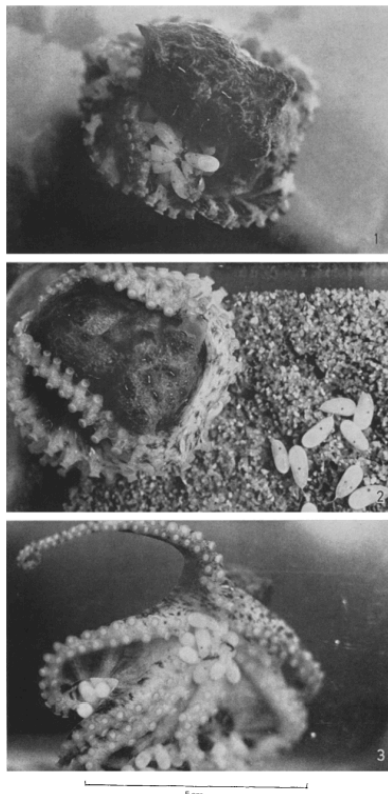


Figure 3: Egg protection by the female *H. maculosa*. **1** et **2**: covering position of *H. maculosa* ; **3**: *H. maculosa* on the move with its eggs (Tranter & Augustine, 1973).

According to Overath and von Boletzky (1974), *H. lunulata* produces small eggs (3.5 mm in diameter) whose para-larvae can be dispersed by ocean currents. This species has the widest range, spanning the central Indo-Pacific. However, *H. maculosa* and *H. fasciata* are holobenthic species, producing larger eggs (between 7 and 9 mm for *H. maculosa* and 6 to 9 mm for *H. fasciata*), which hatch into benthic larvae directly capable of crawling (Tranter & Augustine, 1973 ; Norman et al., 2016 ; Whitelaw et al., 2023 ; Williams & Caldwell, 2024). The eggs of *Hapalochlaena lunulata* take between 4 and 5 weeks to develop, depending on temperature (around 25°C). Higher temperature accelerates development, although other factors, such as nutrient availability, may also influence this process (William & Caldwell, 2024).



Figure 4 : *Hapalochlaena lunulata* egg development. (A) Chromatophores first appear around 1.5 weeks after the start of development. The black dots are the ocelli. (B) The eggs are at 3.5 weeks of development, and hatching is in preparation (William & Caldwell, 2024)
Credit: Roy L. Caldwell.

4.4) Hatching and the first stages of life

H. lunulata juveniles are planktonic at birth, unlike *H. maculosa* juveniles, which are benthic (living close to the seabed) (Williams & Caldwell, 2024). Young blue-ringed octopuses move by propulsion using their arms and can swim close to the water surface. Although mainly planktonic, they can attach themselves to substrates using their suction cups (Overath & von Boletzky, 1974). It appears that young octopuses do not possess the iridophores necessary for forming blue rings; these begin to appear around six weeks post-hatching (Williams & Caldwell, 2024). It has been observed that juveniles flash their blue rings when threatened or during unpleasant situations, with the blue rings used as warning colouration during the early stage of life (Kawashima et al., 2019).

4.5) Maturity

Young octopuses reach maturity in around four months. Once adult, they begin their reproductive cycle. The life cycle of blue-ringed octopuses is very short, with most living no more than a year (Whitelaw et al., 2023 ; Williams & Caldwell, 2024). After reproducing,

both adult males and females enter a phase of senescence, an accelerated aging process. During this period, their health deteriorates rapidly, causing disorientation, tissue damage and physical weakening, leading to death. This final stage is typical of cephalopods, where reproduction marks the end of their life cycle (Lambert, 2011 ; Williams & Caldwell, 2024).

5) Camouflage

5.1) Use of camouflage

Cephalopods skin, especially cuttlefish and octopuses, is highly elastic and muscular, allowing them to give it a tridimensional texture, which they use for communication and camouflage (Mäthger et al., 2012).

Various studies on cephalopod camouflage techniques have highlighted three main contexts in which colour changes occur: homochromy with the environment, emotions such as fear or aggression, and intraspecific interactions (Carlier & Renoue, 2007 ; Mäthger et al., 2012).

In addition to homochromy, these animals also display other colour variations in defensive or agonistic situations. The blue-ringed octopus illustrates this very well. When a predator approaches, the octopus will camouflage itself with its surroundings to remain hidden at first (Hanlon & Messenger, 2018). However if the predator continues to approach, the octopus will display its blue rings in an attempt to intimidate it. In this way, the octopus is no longer hidden but becomes clearly visible to the predator. If the predator continues to approach, the octopus may bite, releasing its toxin (Carlier & Renoue, 2007 ; Hanlon & Messenger, 2018).

More subtle nuances, not always homochromatic, have also been observed in interactions between individuals of the same species. Homochromy thus plays a crucial role in enabling cephalopods to hide effectively from predators while remaining invisible from their prey (Carlier & Renoue, 2007).

5.2) Specialized cells

In order to change the aspect of the skin, cephalopods use a specialized organ called chromatophores. These chromatophores are organs containing a little sac filled with different colors of pigments such as dark brown, black, red and yellow (Carlier & Renou, 2007 ; Mäthger et al., 2009 ; Mäthger et al., 2012). They are surrounded by radial muscle fibers, which are directly controlled by the brain (**Figure 5** ; Mäthger et al., 2009). Chromatophores function as neuromuscular organs rather than simple cells and are not hormone-dependent. By contracting or relaxing, these muscles cause the sac to expand or retract. These chromatophores can expand in less than a second, creating a large range of changeable body patterns, used for communication or camouflage. At maximum expansion, the pigmented area may even disappear completely (Carlier & Renou, 2007; Hanlon et al., 2009 ; Mäthger et al., 2012 ; Messenger, 2001).

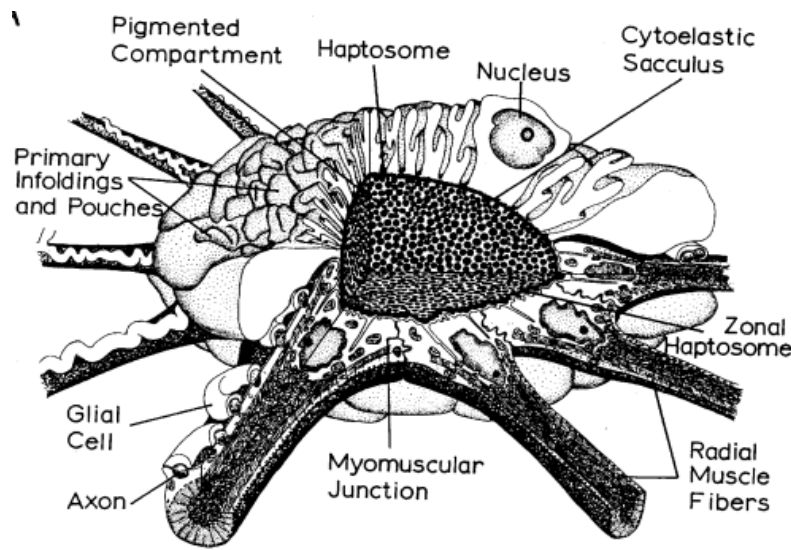


Figure 5 : Diagram of a retracted cephalopod chromatophore (Messenger, 2001).

In order to match perfectly with their environment, they can also use structural reflectors: the leucophores and the iridophores, which are located in layers beneath chromatophores (Mäthger et al., 2009). Leucophores are large cells, responsible for the whiteness in octopus and cuttlefish (Mäthger et al., 2009 ; Mäthger et al., 2012). Iridophores are multilayered reflective cells, made of a protein in between spaces of cytoplasm with a different refractive index. This structure creates interference colors from all the visible spectrum, as well as near UV and infrared light (Carrier & Renoue, 2007 ; Mäthger et al., 2009). When the chromatophores are fully retracted, the leucophores reflect the light optimally, making the animal appear completely white (Messenger, 2001 ; Carrier & Renoue, 2007 ; Mäthger et al., 2012).

5.3) Camouflage control

Cephalopods can adapt their iridescence in two ways : through the muscarinic cholinergic receptors, which can affect the intensity of the iridescence, or through the chromatophores, which will spectrally block or filter the iridescence (Mäthger et al., 2012). The first method can take a few minutes to be effective whereas the second one is almost instant, due to the muscles being under direct neural control (Hanlon & Messenger, 2018). Unfortunately, the pigments in chromatophores usually reflect larger wavelengths (red, orange, yellow, brown) which, by themselves, are insufficient to create strong contrasting patterns and are ineffective for underwater communication at greater depths. That is when structural reflectors, such as iridophores, are useful because they can add shorter wavelength colors. When strong contrast is not needed, structural reflectors are physiologically suppressed or covered by chromatophores (Mäthger et al., 2012 ; Hanlon & Messenger, 2018).

Both strategies have their flaws. The physiological control of iridophores takes time and using chromatophores to mask a reflected signal means that they cannot be used in other body patterns (Mäthger et al., 2012).

5.4) The blue rings

The blue ringed octopus's system, where the iridophores are covered by little modified skin sacs, has yet to be observed in other cephalopods. By hiding their structural reflectors, they make a very visible signal of short- or mid-wavelength colour accessible at all while keeping the chromatophores available to produce other body patterns (Mäthger et al., 2012).

H. lunulata can simultaneously flash its 60 blue rings, which cover the mantle, head and arms, in an aposematic warning display. All its rings can appear in less than 0.3 seconds. (Mäthger et al., 2012). To compare with other octopuses, *Octopus vulgaris* needs two seconds to show a full threat display, with textural, locomotor, chromatic and postural changes on its skin (Mäthger et al., 2012).

When the octopus displays its blue rings, the chromatophores around the rings turn into a dark brown colour, accentuating the contrast and making the blue rings stand out (Hanlon & Messenger, 2018). In addition, the rest of the chromatophores are generally retracted and give a pale appearance, that draws even more attention to the blue rings (**Figure 6** ; Mäthger et al., 2012 ; Hanlon & Messenger, 2018).

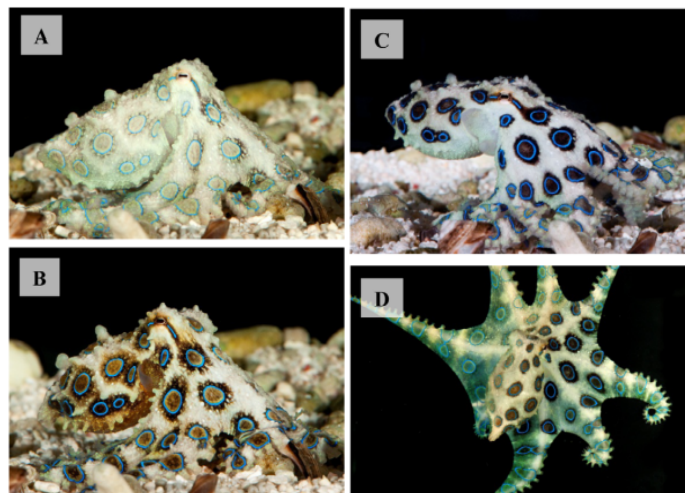


Figure 6: Blue-ringed octopus (*H. lunata*) showing different intensity and contrast of its blue rings. A) Low contrast, absence of brown chromatophores around the rings. B and C) Appearance of chromatophores surrounding the blue rings, increase in contrast. D) Neural control allows unilateral control of the intensity and contrast of the blue rings (Mäthger et al., 2012).

The iridophores responsible for the blue rings are located in modified skin folds forming a pocket, with their opening controlled by the contraction and relaxation of the

muscles located above the iridophore and the ones that are around the chromatophore (**Figure 7** ; Mäthger et al., 2012).

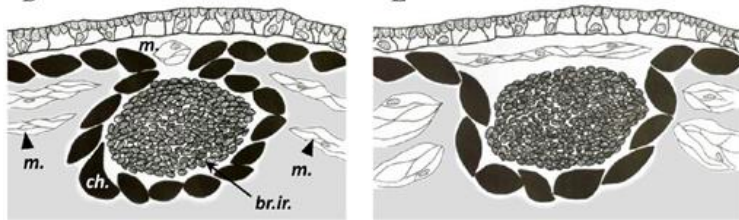


Figure 7: Diagram of the closing (left) and opening (right) of the blue rings (Mäthger et al., 2012).

6) Toxine

6.1) In Cephalopods

Currently, it is currently suggested that all octopods possess a protein venom that allows them to subdue their prey. This venom is produced and stored in the posterior salivary gland (PSG) (Whitelaw et al., 2019 ; Kajino et al., 2024). Some studies highlight the presence of tetrodotoxin (TTX) in the skin and muscles but at lower concentration, especially in the arms and mantle (Kajino et al., 2024 ; Williams & Caldwell, 2024). In comparison, in some species of pufferfishes, the liver, ovary and skin have the highest TTX concentrations, while muscle tissue is often non-toxic (Katikou et al, 2022).

It appears that all the serine proteases present in the PSGs of cephalopods belong only to the cephalopod clade. These serine proteases are the result of an evolutionary convergence present in many venomous lineages of invertebrates and vertebrates (Whitelaw et al., 2020; Whitelaw et al., 2023).

6.2) In *Hapalochlaena*

In blue-ringed octopuses, part of this protein venom has been replaced by a non-protein neurotoxin, the tetrodotoxin (TTX) (Whitelaw et al., 2020; Whitelaw et al., 2023 ; Kajino et al., 2024 ; Williams & Caldwell, 2024). The reduced number and expression of serine proteases in the genus *Hapalochlaena* suggests that the function of their PSG has changed over time. TTX binds to voltage-dependent sodium channels, inhibiting the triggering and propagation of action potentials, which leads to muscle paralysis (Geffeney et al., 2019). In addition to TTX, hyaluronidase has been discovered in *H. masculata*. Hyaluronidase acts on cell membranes by increasing their permeability, and is often used as a dispersal factor, particularly in snake venom (Whitelaw et al., 2020).

In order to store TTX, blue-ringed octopuses have developed a resistance mechanism to prevent self-poisoning. In *H. lunulata*, amino acid substitutions in the loops of sodium channels have been discovered, preventing TTX from binding to the channels. This resistance

has also been observed in organisms that feed on TTX-containing prey and also in TTX-storing organisms such as globefish (Whitelaw et al., 2020).

Sodium channels consist of four homologous domains composed of transmembrane segments. In general, the first two domains are identical across cephalopods. However, in the genus *Hapalochlaena*, regions 3 and 4 have mutations that may be involved in the process of inhibiting TTX-sodium channel binding (Geffeney et al., 2019 ; Katikou et al., 2022). Further research is needed to determine whether these mutations were obtained via a common ancestor or whether they arose independently of each other (Whitelaw et al., 2020).

Resistance similar to that of *Hapalochlaena* to TTX has been observed. For example, saxitoxin (STX) is a cyanotoxin often produced by cyanobacteria and contaminating crustaceans and fish. It has been observed that common octopus can feed on contaminated crustaceans and bivalves without adverse effects. However, no mutation has yet been discovered that could explain this resistance process. It appears that this resistance may be induced by selective pressure due to the ingestion of contaminated prey (Asakawa et al., 2019). Indeed, it has been observed that deep-sea cephalopods not exposed to STX-contaminated prey are not resistant to this toxin. Therefore, it is possible that the mutations enabling resistance to TTX in the genus *Hapalochlaena* were a result of prolonged contact with the toxin (Whitelaw et al., 2020 ; Williams & Caldwell, 2024).

There are two hypotheses concerning the origin of the TTX in *Hapalochlaena* : bioaccumulation after ingestion and production by bacteria living in the host, acquired either via ingestion or endosymbiosis (Jacups & Currie, 2008 ; Whitelaw et al., 2019 ; Whitelaw et al., 2020 ; Kajino et al., 2024). The bacterial origin is favored but there has not been much conclusive evidence. Culturing these bacteria outside of the host might be impossible or secondary inducers from the host might be required to produce or regulate TTX (Whitelaw et al., 2019).

Studies have revealed the presence of TTX-producing bacteria in the posterior salivary glands of octopuses belonging to the genus *Hapalochlaena*. These bacteria include the genus *Pseudomonas*, belonging to the Proteobacteria phylum, which accounts for 41% of all species present in the glands. The next most abundant species is from the *Bacillus* genus, belonging to the *Firmicutes* phylum and representing 21% of the species found in the glands. However, current knowledge and literature do not allow us to understand the TTX biosynthesis pathway in *Hapalochlaena* octopuses (Asakawa et al., 2019 ; Whitelaw et al., 2020).

The highest concentration of TTX are found in the posterior salivary gland and skin of the octopuses *H. lunulata*, *H. fasciata*, and *H. maculosa*. This indicates that TTX serves both hunting (as injected venom) and defense (as ingested poison) against predators (Asakawa et al., 2019 ; Geffeney et al., 2019). In *Hapalochlaena spp.*, the high presence of TTX in the skin suggests that it functions both as poison and venom, and may be transported into the bloodstream through dermal channels for increased efficacy (Williams, 2008 ; Williams & Caldwell, 2024). TTX has also been found in ink, suggesting an additional defensive role (Williams, 2008). *Hapalochlaena* venom can target a wide range of vertebrates and invertebrates. The only octopods-bite induced human death were made by species from *Hapalochlaena* (Whitelaw et al., 2019).

6.3) TTX in other organisms

TTX can be found in both marine and terrestrial species that are phylogenetically unrelated. This toxin has been identified in over 60 species spanning 8 phyla, including Chordata, Echinodermata, Mollusca, Arthropoda, Chaetognatha, Annelida, Platyhelminthes, and Nemertea (Geffeney et al., 2019). The TTX presence in these species has been explained with a bacterial origin. Indeed, commensal associations are then believed to be the source of TTX in many genera, such as pufferfish, gastropods, octopus, and crustaceans. However, these numbers are most likely underestimating the bacterial biodiversity, as not all bacteria can be cultured in laboratory conditions (Whitelaw et al., 2019). These species concentrate their toxins in specific tissues for defense, such as the granular glands of newts or the skin glands of pufferfish (Hanlon & Messenger, 2018).

In nearly all TTX-resistant species, resistance to tetrodotoxin is due to amino acid substitutions in the TTX-binding site of voltage-gated sodium channels (VGSCs), which significantly decrease TTX binding to the channel pore (Geffeney et al., 2019). In the newt, *Taricha granulosa*, the production of TTX has been described as endogenous. TTX is found in the skin, the ovaries and the muscles but no bacteria has been found in these tissues. Studies on this species have shown that they're capable of self-producing TTX when fed a none-TTX diet (Whitelaw et al., 2019). In these cases, TTX-producing bacteria have been isolated, suggesting a bacterial origin for the toxin (Asakawa et al., 2019).

Conclusion

Octopuses of the genus *Hapalochlaena* represent a remarkable example of evolutionary adaptation in the animal kingdom. Their soft bodies, combined with their camouflage abilities allow them to blend perfectly into their environment, both in crevices and on sandy bottoms. In addition, their blue rings produced by iridophores serve as visual signals to scare predators and increase their survival rate. Combined with these visual signals, these octopuses can also use their toxin (tetrodotoxin) as a means of defense and to paralyze their prey. Their life cycles and reproductive strategies, although marked by a short lifespan, effectively ensure the sustainability of the species. The genus *Hapalochlaena* embodies a fine evolution where each morphological, physiological and behavioral trait responds precisely to the requirements of their environment.

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